AI-103 Study Guide: Module 1.2 Enhanced

Select, Deploy, and Evaluate Microsoft Foundry Models

This is an ENHANCED version with critical benchmark details filled in from TXT lecture

1. Exam Domain Mapping

AI-103 Exam Domain	Relevance	Why
Plan and manage an Azure AI solution	**High**	Covers model selection, deployment options, endpoints, cost, quota, performance and evaluation planning.
Implement generative AI and agentic solutions	**High**	Covers choosing LLMs/SLMs, deploying models, using playgrounds, testing prompts and evaluating generative output.
Implement information extraction solutions	Medium	Embedding models and RAG readiness support retrieval and grounding scenarios.
Implement text analysis solutions	Low–Medium	Evaluation metrics such as fluency, coherence, relevance and NLP metrics support text output assessment.
Implement computer vision solutions	Low	The catalog includes image, video and multimodal models, but this module focuses mainly on selection/deployment/evaluation.

Exam focus: Choosing the right model, deploying it correctly, testing it, and proving it meets quality, safety, cost and performance requirements.

2. What This Module Teaches

This module teaches the end-to-end workflow for working with models in Microsoft Foundry:

1. Explore the model catalog
2. Filter and compare models
3. Use benchmarks to make selection decisions
4. Deploy the selected model to an endpoint
5. Test the model in the playground
6. Evaluate model performance using manual and automated approaches
7. Iterate using prompt engineering, RAG, fine-tuning, model changes or safety controls

Key exam idea:

> *Do not select a model based only on popularity or size. Select it based on task fit, quality, safety, cost, throughput, latency, and required features.*

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3. Key Concepts

3.1 Model Catalog

The **Microsoft Foundry model catalog** is the central place to discover, compare and deploy models.

Catalog Feature	Purpose
Search	Find models by name or keyword
Filters	Narrow models by collection, capability, provider, inference task, industry or fine-tuning support
Model cards	Show provider, capabilities, benchmarks, responsible AI notes and deployment options
Leaderboard	Compare models by quality, safety, cost and performance
Trade-off charts	Compare two dimensions, such as quality vs cost or quality vs throughput
Side-by-side comparison	Compare two or three models across benchmarks, features, endpoints and context

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3.2 Model Types

Model Type	Use For	Trade-off
Large Language Models (LLMs)	Complex reasoning, deep content generation, long context, sophisticated chat	Higher cost and compute
Small Language Models (SLMs)	Faster, cheaper, simpler NLP tasks, edge/lower compute scenarios	Less capable for complex reasoning
Reasoning models	Complex problem solving, coding, math, science, strategy, logistics	May be slower or more expensive

Model Type	Use For	Trade-off
Embedding models	Semantic search, recommendations, RAG, similarity matching	Not used for direct chat responses
Image generation models	Generate images from prompts	Vision generation scenarios
Video generation models	Generate video from prompts	Video generation scenarios
Image analysis / multimodal models	Interpret images with text prompts	Vision understanding scenarios
Text-to-speech models	Convert text to audio	Speech output
Speech-to-text models	Convert speech/audio to text	Transcription and voice input
Domain-specific models	Specialist industry/language tasks	Better when domain accuracy matters

3.3 Benchmarks (CRITICAL FOR EXAM)

Benchmarks help compare models objectively.

Quality Benchmarks

Quality benchmarks assess how well a model generates accurate, coherent, and contextually appropriate responses.

Quality Metric	What It Measures
Quality Index	Overall quality score (0-1) averaging accuracy across multiple benchmark datasets
Accuracy	Correctness of model output
Coherence	Logical flow and consistency of responses
Fluency	Linguistic correctness and natural language quality
Relevance	Whether the answer addresses the user's question/request
Groundedness	Whether the answer is based on provided context rather than speculation

Quality Index Datasets Used:

* **Arena-Hard** - Adversarial question answering
* **BIG-Bench Hard** - Reasoning capabilities

* **GPQA** - Graduate-level multi-discipline questions
* **HumanEval+** - Code generation tasks
* **MBPP+** - Code generation tasks
* **MATH** - Mathematical reasoning
* **MMLU-Pro** - General knowledge assessment
* **IFEval** - Instruction following

Exam memory: If the question asks about linguistic correctness or natural language quality, the answer is **fluency**.

Safety Benchmarks

Safety metrics ensure models don't generate harmful, biased, or inappropriate content.

Safety Area	Benchmark	What It Checks
Harmful behavior	**HarmBench**	Cybercrime, illegal activities, general harm, misinformation, harassment, bullying, copyright violations. Attack Success Rate (ASR) - lower is safer.
Toxic content	**ToxiGen**	Adversarial and implicit hate speech, toxicity. F1 scores for detection across references to minority groups.
Protected material	—	Copyright or proprietary material reproduction risk
Sensitive domain knowledge	**WMDP**	Weapons of Mass Destruction Proxy - measures model knowledge in biosecurity, cybersecurity, chemical security. Higher scores = more dangerous knowledge.
Indirect attack / jailbreak	—	Vulnerability to manipulation or prompt injection attempts

Cost Benchmarks

Cost Metric	Meaning
Input token cost	Cost to process 1 million input tokens (text you send to model)
Output token cost	Cost to generate 1 million output tokens (text model produces)
Estimated cost	Combined cost estimate using typical 3:1 ratio (three input tokens per output token)

Exam memory: A larger or higher-quality model may not be the best choice if a cheaper/faster model satisfies the requirement.

Performance Benchmarks

Performance Metric	Meaning
Latency mean	Average response time in seconds

Performance Metric	Meaning
Latency P50/P90/P95/P99	Percentile-based response times (50%, 90%, 95%, 99% of requests complete within this time)
Time to first token (TTFT)	How quickly streaming output starts
Generated tokens per second (GTPS)	Output generation speed
Total tokens per second (TTPS)	Overall token processing throughput
Time between tokens	Delay between streamed tokens

Exam memory: If the question asks about processing prompts and returning comprehensive responses quickly, think **throughput**.

3.4 Deployment Types

Deployment Type	Data Residency	Billing	Best For	Quota
Global Standard	Any region	Pay-per-token	General workloads	Largest
Global Provisioned	Any region	Reserved PTUs	High-throughput workloads	—
Global Batch	Any region	50% discount	Large async jobs (24h)	—
Data Zone Standard	EU/US zone	Pay-per-token	Compliance (data zone)	—
Data Zone Provisioned	EU/US zone	Reserved PTUs	High-throughput + compliance	—
Data Zone Batch	EU/US zone	Discounted	Large async + compliance	—
Standard	Single region	Pay-per-token	Regional residency or low volume	—
Regional Provisioned	Single region	Reserved PTUs	Regional + high throughput	—
Developer	Any region	Pay-per-token	Fine-tuned model evaluation only	—

Exam memory: For general use with largest quota, choose **Global Standard**.

3.5 Evaluation Metrics

Generation Quality Metrics (AI-Assisted)

Metric	What It Measures	Use When
Groundedness	Whether responses are based on provided context (not speculation)	Need factual accuracy (e.g., customer support)
Relevance	Whether responses address the user's question appropriately	Any generative task
Coherence	Whether responses flow logically and maintain consistent ideas	Narrative or explanation tasks
Fluency	Linguistic correctness and natural language quality	Need grammatically correct output

Safety Metrics (Automated Classification)

Risk Category	What's Detected
Self-harm content	Responses discussing or encouraging self-harm
Hateful and unfair content	Bias, discrimination, or hateful statements
Violent content	Responses containing or promoting violence
Sexual content	Inappropriate sexual content
Protected material	Copyright or proprietary content reproduction
Indirect attack (jailbreak)	Vulnerability to manipulation attempts

Defect rate = (instances exceeding severity threshold / total instances) × 100

NLP Metrics (Mathematical - need ground truth)

Metric	Best For	Calculation
F1-score	Text classification, information retrieval	Balances precision and recall; ratio of shared words
BLEU	Machine translation evaluation	Compares n-grams (word sequences)
METEOR	Translation evaluation	Extends BLEU by accounting for synonyms, stemming, paraphrasing
ROUGE	Summarization tasks	Emphasizes recall over precision; covers key points
GLEU	Sentence-level evaluation	Variant of BLEU designed for sentence-level comparison

Exam memory: Use NLP metrics when you have definitive correct answers or reference texts. Use AI-assisted metrics for open-ended generation.

4. Azure Services / Tools

Service / Tool	Role	Exam Relevance
Microsoft Foundry portal	Explore catalog, compare models, deploy, test and evaluate	High
Foundry Models catalog	Central repository of 1,900+ models	High
Model cards	Show capabilities, benchmarks, responsible AI information	High
Model leaderboard	Compare models by quality, safety, cost and throughput	High
Trade-off charts	Compare dimensions such as quality vs cost	High
Model playground	Manual testing and prompt experimentation without code	High
Evaluation feature	Run systematic model/agent/dataset evaluations	High
Evaluator library	Manage Microsoft-curated and custom evaluators	Medium
Project endpoint	Access Foundry models and Foundry-specific functionality	High
OpenAI v1 / Azure OpenAI endpoint	OpenAI-compatible API access	High
Microsoft Entra ID	Recommended authentication for production scenarios	High

Service / Tool	Role	Exam Relevance
Azure AI Content Safety	Additional protection for harmful input/output	Medium–High
Azure Monitor / Application Insights	Monitor production usage, latency, cost and errors	Medium–High

5. Decision Rules

Scenario	Best Choice	Why
Need highest quality for complex reasoning	LLM or reasoning model	Better reasoning and deeper context handling
Need lower cost and faster response for simpler tasks	SLM	Efficient and cheaper for common NLP tasks
Need semantic search or RAG	Embedding model	Converts text into vectors for similarity search
Need text/image understanding	Multimodal model	Accepts image and text input
Need text-to-image output	Image generation model	Purpose-built for image generation
Need speech transcription	Speech-to-text model	Converts audio into text
Need voice response	Text-to-speech model	Converts text into audio
Need compare model suitability	Leaderboard + model card + benchmarks	Gives objective comparison
Need balance quality and budget	Quality vs cost trade-off chart	Helps avoid overpaying for unnecessary capability
Need real-time chat responsiveness	Low latency + high throughput model	Better interactive user experience
Need high-volume predictable throughput	Provisioned deployment	Reserved throughput gives predictability
Need general use and largest quota	Global Standard deployment	Best general-purpose choice with largest quota
Need EU/US data zone compliance	Data Zone deployment	Keeps data within required zone
Need large asynchronous low-cost processing	Batch deployment	Suited for non-interactive jobs
Need production authentication	Microsoft Entra ID	More secure than static keys

Scenario	Best Choice	Why
Need quick prompt testing	Playground	No-code testing and comparison
Need systematic quality assessment	Evaluation run with dataset and metrics	Scalable, repeatable evaluation
Evaluation scores are weak	Prompt engineering, model change, RAG or fine-tuning	Improve quality based on failure cause
Safety scores are weak	Content filters, prompt hardening, output validation	Reduce harmful or unsafe responses

6. Exam Traps

Trap	Correct Understanding
Bigger model is always better	Not always. Consider task fit, cost, latency, throughput and safety
Quality index is the only benchmark that matters	Safety, cost and performance may be more important depending on scenario
SLMs are inferior	SLMs are often better for cost-sensitive, fast or simpler tasks
Embedding models generate chat responses	Embedding models create vectors for search/RAG, not natural-language answers
Playground testing is enough for production	Playground is useful for manual testing, but production needs systematic evaluation and monitoring
Keys are best for production authentication	Microsoft Entra ID is recommended for production
Global Standard is always wrong because it may use any region	Global Standard is best for general use and largest quota when data residency does not restrict it
Data Zone deployment is for performance only	It is mainly about data zone compliance (EU/US data residency)
Batch deployment is good for interactive chat	Batch is for large asynchronous jobs, not real-time interaction
Fluency measures factual correctness	Fluency measures linguistic correctness and natural language quality
Groundedness measures grammar	Groundedness measures whether responses are based on supplied context
Relevance and groundedness are the same	Relevance checks if answer addresses question; groundedness checks if it's supported by context
Safety benchmark means the app is fully safe	You may still need content filters, prompt hardening, output validation and monitoring

Trap	Correct Understanding
Fine-tuning is always the next step after poor results	First consider prompt engineering, model choice or RAG; fine-tuning can add complexity and cost

7. Hands-on Checklist

* [ ] Open the Microsoft Foundry model catalog
* [ ] Search and filter models by capability, provider, task and deployment option
* [ ] Read a model card and identify capabilities, benchmarks and deployment options
* [ ] Use the model leaderboard and sort by different benchmarks
* [ ] Compare models using quality, safety, cost and throughput trade-off charts
* [ ] Compare two models side by side in detail
* [ ] Deploy a model using default settings
* [ ] Identify the deployment name for programmatic access
* [ ] Find the endpoint URL and authentication details
* [ ] Test a deployed model in the playground
* [ ] Modify system messages and parameters in the playground
* [ ] Compare model outputs side by side in the playground
* [ ] Create an evaluation run with a test dataset
* [ ] Use synthetic data generation for evaluation
* [ ] Select evaluation criteria (quality, safety, NLP metrics)
* [ ] Review evaluation results and identify failures
* [ ] Identify whether to improve through prompts, model choice, RAG, fine-tuning or safety controls
* [ ] Delete lab resources to avoid unnecessary cost

8. Quick Revision Questions

1. What is the purpose of the Microsoft Foundry model catalog?
2. What information is available on a model card?
3. When would you choose an SLM instead of an LLM?
4. When would you use an embedding model?
5. What are the four main benchmark categories used to compare models?
6. What does the quality index measure?
7. Which benchmark helps you assess harmful content risk?
8. Which metric indicates how quickly a model can process and return responses? (Answer: **Throughput**)
9. What does time to first token (TTFT) measure?
10. What deployment type is best for general use and largest quota? (Answer: **Global Standard**)
11. When would you choose a Data Zone deployment?
12. When would you choose a Batch deployment?
13. What three details does an application need to call a deployed model?
14. Why is Microsoft Entra ID preferred over keys for production?

15. What is the purpose of the model playground?
16. What does groundedness measure?
17. What does fluency measure?
18. What is the difference between relevance and coherence?
19. When are NLP metrics like ROUGE or BLEU useful?
20. What should you do if evaluation results show low quality?
21. What should you do if safety metrics show concerns?
22. Why should evaluation be repeated after changes?
23. Why is model selection a trade-off decision rather than a single-score decision?

Final Exam Memory

Remember this sequence:

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Select → Compare → Deploy → Test → Evaluate → Improve → Monitor
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And remember the trade-off:

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Best model = task fit + quality + safety + cost + performance + compliance + deployment support
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For AI-103, **model selection is not just about choosing the most powerful model**. It is about choosing the model and deployment approach that best fits the scenario.

Key Enhancements in This Version

OK Added benchmark dataset names (Arena-Hard, GPQA, MATH, etc.)
OK Added safety benchmark details (HarmBench, ToxiGen, WMDP)
OK Added comprehensive benchmark table with what each measures
OK Added all 9 deployment types with data residency details
OK Added evaluation metrics matrix (groundedness, fluency, coherence, relevance)
OK Added NLP metrics with use cases (ROUGE, BLEU, METEOR)
OK Added quality vs safety trade-off distinctions
OK Clarified exam traps with specific metrics